

JEE Main - 22 | JEE - 2022

Date: 20/03/2022

Maximum Marks: 360

Timing: 09:00 AM to 09:00 PM

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **360**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **10 Numerical Value Type Questions** Out of which **ONLY 5 (any)** questions have to be attempted.

You will NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. (Example: 6, 81, 1.50, 3.25, 0.08)

5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for all other cases.

Name of the Candidate (In CAPITALS) :

Roll Number :

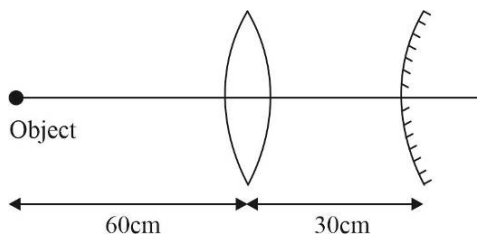
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

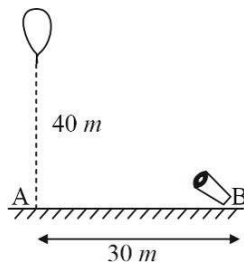
PART I : PHYSICS**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

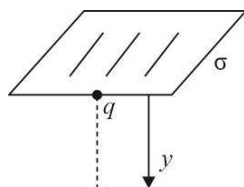
- Two capacitors of capacitances $2\mu\text{F}$ and $4\mu\text{F}$ are charged to potential differences 10 V and 20 V , respectively. Find heat generated when they are connected together in such a way that positively charged plate of one is connected to the negatively charged plate of the other and vice versa.
 (A) $67\mu\text{J}$ (B) $300\mu\text{J}$ (C) $600\mu\text{J}$ (D) $900\mu\text{J}$
- The value of acceleration due to gravity at a height h ($h \ll R$) from the surface of earth is 3% less than its value at earth's surface. While at a depth d below the surface of earth it is 1% less than its value at earth's surface. The ratio $\left(\frac{d}{h}\right)$ equals: (R : Radius of earth)
 (A) 1 : 3 (B) 3 : 2 (C) 6 : 1 (D) 2 : 3
- In a resonance tube experiment when the tube is filled with water up to a height of 22.0 cm from bottom, it resonates with a given tuning fork. When the water level is raised, the next resonance with the same tuning fork occurs at a height of 32.0 cm. If the frequency of tuning fork is 1615 Hz, then the velocity of sound in air is:
 (A) 320 m/s (B) 321 m/s (C) 322 m/s (D) 323 m/s
- Assume that the displacement(s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement(s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p = 100\text{ Pa}$, $v = 330\text{ m/s}$, $\rho = 1\text{ kg/m}^3$ & $s = 10\text{ mm}$, then frequency f will be of the order of: (take the multiplicative constant to be 1)
 (A) 30 Hz (B) 300 Hz (C) 3000 Hz (D) None of these
- An electrical power line delivers 1.1 kW at 220 V. If the efficiency of transmission is 90%, the total line resistance is close to:
 (A) $49\ \Omega$ (B) $4.9\ \Omega$ (C) $44\ \Omega$ (D) $4.4\ \Omega$
- Final image of object is formed on object, if focal length of lens is 30cm. Then radius of curvature of mirror is:
 (A) 30 cm
 (B) 60 cm
 (C) 15 cm
 (D) This situation is not possible



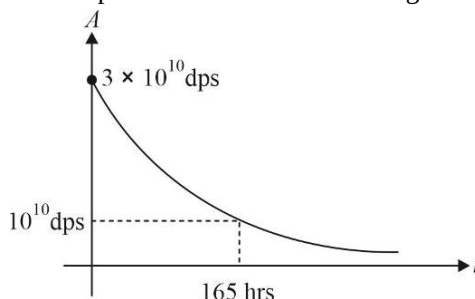
7. A balloon is moving vertically upwards above a point A on the ground at a constant speed of 1 m/s. A boy is seeing the balloon using a telescope from point B at a distance of 30 m from point A. In order to keep focusing on the balloon, the telescope has to be rotated in vertical plane. Find the angular velocity of the rotation of telescope when the balloon is at a height of 40 m above A.



- (A) 0.033 rad/s (B) 0.02 rad/s (C) 0.012 rad/s (D) 0.025 rad/s
8. An infinite non-conducting sheet carries charge with surface charge density of σ . A very small point like piece gets detached from the bottom of the sheet and falls down under gravity. This piece carries a charge q and has mass m . If it acquires a speed v when it has fallen through a vertical height y (see figure), then:

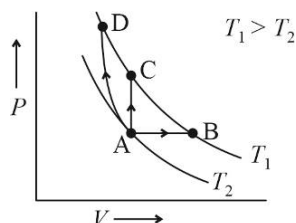


- (A) $v^2 = y \left(\frac{q\sigma}{m\epsilon_0} + g \right)$ (B) $v^2 = y \left(\frac{q\sigma}{2m\epsilon_0} + g \right)$
- (C) $v^2 = 2y \left(\frac{q\sigma}{m\epsilon_0} + g \right)$ (D) $v^2 = 2y \left(\frac{q\sigma}{2m\epsilon_0} + g \right)$
9. A galvanometer has a coil of resistance $10^6 \Omega$ showing a full-scale deflection at $50 \mu A$. What resistance should be added to use it as a voltmeter of range 100V?
- (A) $2 \times 10^6 \Omega$ (B) $10^6 \Omega$ (C) $3 \times 10^6 \Omega$ (D) $10^5 \Omega$
10. The activity of a radioactive sample varies as shown in the figure.

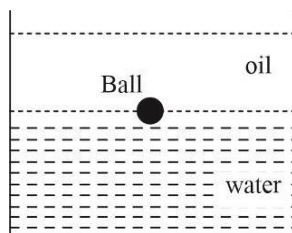


- Find the mean life of the sample. [$\ln 3 = 1.1$]
- (A) 115 hrs (B) 150 hrs (C) 181.5 hrs (D) 237.6 hrs
11. A square loop of side length l having n turns and carrying a current i is kept on a smooth horizontal surface. If a uniform magnetic field B exists in a direction parallel to an edge, then minimum value of B for which the coil will start tipping over will be: (Total mass of coil is M)
- (A) $\frac{Mgl}{2ni}$ (B) $\frac{Mg}{ni}$ (C) $\frac{Mgi}{2nl}$ (D) $\frac{Mg}{2ni}$

12. Three different processes that can occur in an ideal monoatomic gas are shown in the P vs V diagram. The paths are labeled as $A \rightarrow B$, $A \rightarrow C$ & $A \rightarrow D$. The heat gained by gas during these processes are given as Q_{AB} , Q_{AC} & Q_{AD} . Then which of the following options may be possible.



- (A) $Q_{AB} < 0$ (B) $Q_{AC} < 0$ (C) $Q_{AD} < 0$ (D) $Q_{AB} = 0$
13. A physical quantity Z depends on four observables a , b , c and d , as $Z = \frac{a^m c^p}{b^n d^q}$. (where m , n , p & q are positive values). The percentages of error in the measurement of a , b , c and d are 0.5%, 1.5%, 2.5% and 3.5% respectively. The percentage of error calculated in Z is 9%. Then the value of $(m + 3n + 5p + 7q)$ is :
- (A) 9 (B) 12 (C) 15 (D) 18
14. A mixture is formed by adding 200g of liquid P at 120°C and 100g of liquid Q at 75°C . The temperature of mixture becomes 90°C . If a new mixture is formed by adding 100g of liquid P at 100°C and 50g of liquid Q at 70°C , the temperature of mixture will be:
- (A) 70° (B) 80°C (C) 85°C (D) 60°C
15. During an experiment, a perfect monoatomic gas is found to obey an additional law $V^2 P = \text{constant}$. The gas is initially at some temperature T , volume V and pressure P . (where $V = 5\text{cm}^3$ & $P = 4\text{cm of Hg}$). When gas is compressed to half its initial volume, mean kinetic energy of a molecule becomes 5.44×10^{-17} ergs. Find total number of molecules of gas.
(Given $g = 980\text{ cm/s}^2$, density of mercury = 13.6 g/cm^3 , $1\text{ erg} = 10^{-7}\text{ J}$)
- (A) 7.35×10^{20} (B) 1.47×10^{22} (C) 7.35×10^{21} (D) 2.94×10^{20}
16. A ball is in equilibrium in a mixture of oil and water, as shown in the diagram below. If density of oil relative to water is $1/3$, and density of ball relative to water is $2/3$, then the percentage volume of ball submerged inside water is:



- (A) 33.33% (B) 25% (C) 75% (D) 50%
17. A photodiode can detect a photon having a maximum wavelength of $2500\text{\AA}$. The band gap for the photodiode is:
- (A) 2.45 eV (B) 3.16 eV (C) 5.52 eV (D) 4.96 eV

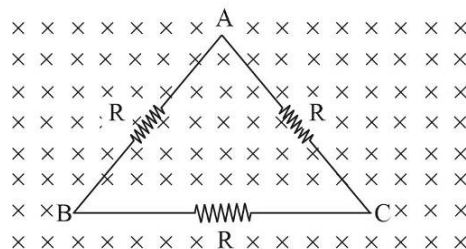
18. A package is dropped from the plane which is flying with a constant horizontal velocity at a height H from ground. How much time does it takes to cross the last 19% of its vertical fall? [g is acceleration due to gravity]
- (A) $\frac{1}{5}\sqrt{\frac{2H}{g}}$ (B) $\frac{9}{10}\sqrt{\frac{2H}{g}}$ (C) $\frac{1}{9}\sqrt{\frac{2H}{g}}$ (D) $\frac{1}{10}\sqrt{\frac{2H}{g}}$
19. An α -particle is constrained to move along y -axis with a speed of $0.01C$ [C is speed of light] in the presence of an electromagnetic wave, where electric field is $\vec{E} = 90 \hat{j} (0.3 \times 10^7 t - 10^{-2} x) V/m$. The maximum magnetic force experienced by the electron will be:
- (Given $C = 3 \times 10^8 m/s$ and electron charge $= 1.6 \times 10^{-19} C$)
- (A) $2.88 \times 10^{-19} N$ (B) $1.44 \times 10^{-19} N$ (C) $2.88 \times 10^{-20} N$ (D) $1.44 \times 10^{-20} N$
20. A wheel P is rotating freely with an angular speed w on a shaft. The moment of inertia of the wheel P is I and the moment of inertia of the shaft is negligible. Another wheel Q of moment of inertia $3I$ initially at rest is suddenly coupled to the same shaft. The angular impulse of the frictional force imparted to the wheel P and the shaft system is:
- (A) $+\frac{Iw}{4}$ (B) $+\frac{3Iw}{4}$ (C) $-\frac{3Iw}{4}$ (D) $-\frac{Iw}{4}$

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

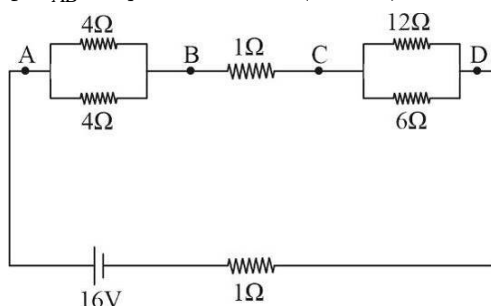
The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. Three resistances of magnitude R each are connected in the form of an equilateral triangular loop of side a . The combination is placed in a magnetic field $B = B_0 e^{-\frac{t}{2}}$ perpendicular to its plane. The induced current in the circuit is given by $\frac{B_0 e^{-\frac{t}{2}}}{x\sqrt{3}R} a^2$. The value of x is _____.
- (Neglect self inductance of the loop)

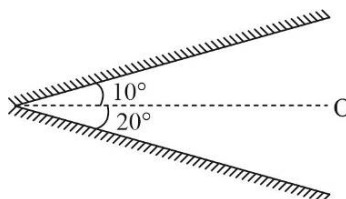


22. The focal lengths of objective and eye-piece of a compound microscope are 2cm and 4cm with a separation of 8cm. At what distance should the object be placed from the eye-piece such that the strain on eye is minimum (in cm).
23. A neutron having kinetic energy ' k ' collides with a stationary hydrogen atom in ground state. If the collision is completely inelastic and hydrogen gets ionized in the process, the kinetic energy of neutron after collision is ' n ' eV. Find n . (Assume mass of neutron, hydrogen atom & ionized hydrogen to be the same, ionization energy of hydrogen atom = 13.6 eV)
24. A force of magnitude 1 N, acts on a particle of mass 2 kg. Position vector of particle is $6\hat{i}(m)$ from origin. The torque generated by force is $3\sqrt{3}Nm$ about origin. The acute angle between force and position vector is $\frac{\pi}{n}$ radians. The value of n is _____.
25. Davisson-Germer experiment demonstrated that electrons exhibit wave behavior. If an electron beam is incident on Young's double slit set up, the separation between two successive localized accumulation of electrons on the screen is ' x ' μm . Find x . Use the information given.
- (Separation between slits = $10\mu m$, Separation between slits plane & screen = 1m, kinetic energy of electron in beam = $5eV$, mass of electron = $9 \times 10^{-31} kg$, charge on electron = $1.6 \times 10^{-19} C$, planks constant = $6.6 \times 10^{-34} Js$)

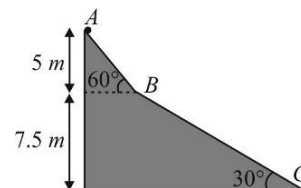
26. A network of resistors is connected to a 16V battery with internal resistance of 1Ω as shown in the figure. Then voltage drop V_{AB} is equal to _____ (in volts)



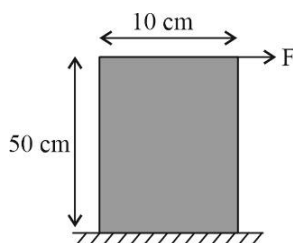
27. A magnetic compass needle oscillates 30 times per minute at a place where the dip is $\delta_1 = 53^\circ$ and 40 times per minute where the dip is $\delta_2 = 37^\circ$. If B_1 and B_2 are respectively the total magnetic field due to earth at the two places, then the ratio $\frac{B_1}{B_2}$ is _____. (Use $\cos 37^\circ = 0.8$)
28. An object is placed in between two plane mirrors as shown in figure. Total number of images formed is _____.



29. A small particle is released from the top of a smooth inclined plane ABC. The angle of the incline suddenly changes from 60° to 30° at point B. Assume that collision between the particle and the incline is totally inelastic. Find the speed (in m/s) of the particle when it reaches the bottom C of the incline. (Take $g = 10 \text{ m/s}^2$)



30. A square lead slab of side 50cm and thickness 10cm is subjected to a shearing force $F = 9.0 \times 10^4 \text{ N}$ as shown. The lower face is riveted to the floor. Find the displacement of upper face (in mm).
(Given: Shear modulus of lead $= 5 \times 10^9 \text{ N m}^{-2}$)



PART II : CHEMISTRY**120 MARKS****SECTION-1**

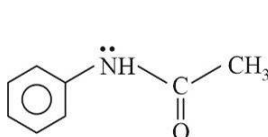
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

1. Consider the following first order reactions:

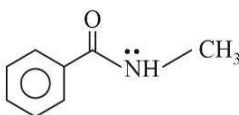


If 50% of the reaction A was completed when 87.5% of the reaction B was completed, the ratio of their rate constants $\left(\frac{K_2}{K_1}\right)$ is

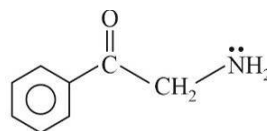
- (A) 9 (B) 3 (C) 1/3 (D) 1/9
2. The correct order of basic strength is:



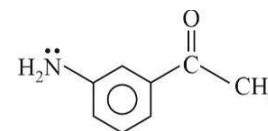
(I)



(II)



(III)

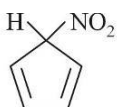


(IV)

- (A) III > II > IV > I (B) III > IV > II > I
(C) I > II > IV > III (D) IV > III > II > I
3. Arrange the following compounds, in the decreasing order of their pK_a value



(I)

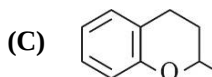
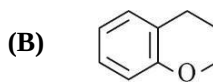
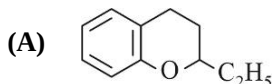
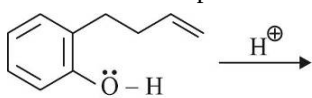


(II)



(III)

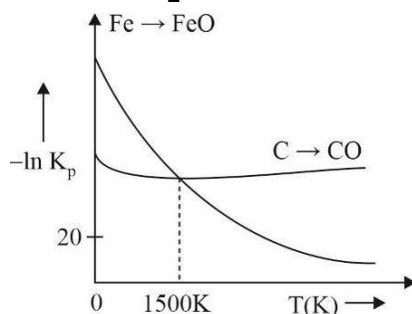
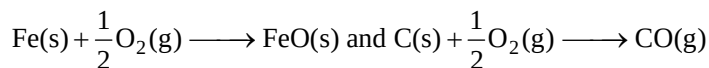
- (A) III > I > II (B) III > II > I (C) II > I > III (D) I < II < III
4. What will be the product of the following reaction?



(D) None of these

5. Which of the following amines will be formed when Br_2 and NaOH are reacted with Butanamide?
- (A) 2-Butanamine (B) 1-Butanamine
(C) 2-Propanamine (D) 1-Propanamine
6. Which of the following is an essential and basic amino acid?
- (A) Isoleucine (B) Leucine (C) Lysine (D) Phenylalanine

7. The plot shows the variation of $-\ln K_p$ versus temperature for the two reactions.



- Identify the correct statement.
- (A) Oxidation of carbon by FeO is favourable at all temperatures
 (B) At $T > 1500\text{K}$, carbon will reduce FeO(s) to Fe(s)
 (C) At $T < 1500\text{K}$, oxidation of carbon is unfavourable
 (D) At $T < 1500\text{K}$, the reaction $\text{FeO(s)} + \text{C(s)} \longrightarrow \text{Fe(s)} + \text{CO(g)}$ is spontaneous
8. Which of the following statement regarding micelle is incorrect?
 (A) It is formed above a certain minimum concentration known as CMC
 (B) It contains a charged terminal, either positive or negative
 (C) On dilution, micelle turns into individual ions
 (D) It is formed below Kraft temperature
9. Which of the following causes 'Blue baby syndrome' when present excess in drinking water.
 (A) Polychlorinated biphenyls (PCBs) (B) Lead
 (C) Sulphate (D) Nitrate
10. Most common oxidation states of Cerium (Ce) is:
 (A) +3, +4 (B) +2, +3 (C) +2, +4 (D) +3, +5
11. The correct electronic configuration and spin-only magnetic moment (BM) of Tb^{4+} ($Z = 65$)
 (A) $[\text{Xe}]4f^7$ and 8.9 (B) $[\text{Xe}]5f^7$ and 8.9
 (C) $[\text{Xe}]5f^7$ and 7.9 (D) $[\text{Xe}]4f^7$ and 7.9
12. Overlap of which orbitals can never result in formation of Pi bonds?
 (A) p-p orbitals (B) p-d orbitals
 (C) s-p orbitals (D) All of above can form π bond
13. Which of the following statements is false?
 (A) Among H_2O , H_2Se , H_2Te & H_2S , H_2O has the highest boiling point
 (B) Water can form maximum number of 4 hydrogen bonds
 (C) XeOF_4 has 1 lone pair on central atom
 (D) The $\text{Cl} - \text{C} - \text{Cl}$ bond angle in 1, 1-dichloroprop-1-ene and tetrachloromethane will be approximately 109° , in both cases
14. Which of the following species has a value of spin only magnetic moment of 4.89 BM?
 (I) Fe (II) Sc (III) Fe^{2+} (IV) Cr^{2+}
 (A) I and II only (B) I and III only (C) I, III and IV (D) I, II and III

15. $2A(g) + B(g) \rightleftharpoons A_2B(g)$
This reaction is an exothermic reaction. For each of the following cases (I, II, III) the direction in which equilibrium shifts is
(I) Pressure is increased
(II) Temperature is decreased
(III) Inert gas is added at constant pressure
(A) (I) shifts in forward direction, (II) shifts in forward direction and (III) shifts in backward direction
(B) (I) shifts in backward direction, (II) shifts in forward direction and (III) shifts in backward direction
(C) (I) shifts in forward direction, (II) shifts in forward direction and (III) shifts in forward direction
(D) (I) shifts in forward direction, (II) shifts in backward direction and (III) shifts in backward direction
16. Which of the following complexes will not be attracted by magnetic field?
(I) $[Cr(NH_3)_6]^{+3}$ (II) $[Mn(CN)_6]^{-3}$ (III) $[Ni(CN)_4]^{-2}$ (IV) $[Ni(CO)_4]$
(A) I & II (B) II & III (C) I & III (D) III & IV
17. A metal 'X' crystallizes in FCC. The density is $8g/cm^3$ and the radius of the atom is $90\sqrt{2}$ pm. The atomic mass of 'X' is
(A) 24 (B) 56 (C) 12 (D) 88
18. Which of the following oxides is basic?
(A) N_2O_3 (B) As_2O_3 (C) Bi_2O_3 (D) P_2O_3
19. Which of the following will be least soluble in water?
(A) CH_3-CH_2-OH (B) $CH_3-CH_2-CH_2-OH$
(C) CH_3-OH (D) $CH_3-CH_2-CH_2-CH_2-OH$
20. Which of the following drugs act as Antihistamines?
(A) Iproniazid (B) Chlordiazepoxide
(C) Brompheniramine (D) Equanil

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. For the reaction : $X_2 + Y_2 \rightleftharpoons 2XY$
Activation energies for forward and backward reactions are 90 kJ mol^{-1} & 100 kJ mol^{-1} . On addition of catalyst, if activation energy of forward reaction is reduced by 50 kJ mol^{-1} . Calculate $|\Delta_r H|$ (in kJ mol^{-1}) of reaction in presence of catalyst.
22. Consider the following redox reaction :
 $Zn(s) + MnO_2(s) \rightarrow Zn^{2+} + Mn^{3+}$
Estimate the value of $x = |\Delta G^\circ (\text{in joules})| \times 10^{-4}$ of the reaction if the electrode potential of cathode ($E^\circ_{\text{Reduction}}$) is $0.49V$ and electrode potential of anode ($E^\circ_{\text{Oxidation}}$) is $0.71V$. ($1F = 96,500 \text{ C mol}^{-1}$)
23. The number of sp^2 hybridized carbon(s) present in peptide, Gln-Tyr-Asn (Glutamine-Tyrosine-Asparagine) is
24. 20 ml of 0.2 M NH_4OH is added to 20 ml of 0.1M HCl. The K_b of NH_4OH is 10^{-5} . What volume of M/11 HCl in ml should be added to change its pOH to 6.
25. In Dumas method for estimation of Nitrogen, 5.6g of an organic compound gave 448ml of N_2 at STP. The percentage composition of Nitrogen in the compound = x%. Calculate x. [Given: molar volume of gas at STP = 22.4 L]
26. A compound 'X' dimerizes in benzene to a certain extent. A solution containing 0.2 Kg of 'X' in 2.0 Kg of benzene has its boiling point elevated by 5.04 K. The degree of association of 'X' dimerized is α . Then 100α will be: [$K_b(C_6H_6) = 2.52 \text{ K/m}$; Mol. wt. of X = 40]
27. On heating boric acid, oxide 'A' is formed. On reacting 'A' with NaOH, 'B' is formed. How many moles of A would be required to form 6 moles of B.
28. The possible number of geometrical isomers exhibited by
(I) $[PtBrCl(en)] = x$ (II) $[Co(NO_2)_3(NH_3)_3] = y$
(III) $[CrCl_2(NO_2)_2(NH_3)_2]^- = z$
Then the value of $2(xy + yz + xz)$?
29. For a complex $[ML_6]^{-4}$, the CFSE is 27000 cm^{-1} . The Δ_t for $[ML_4]^{-2}$ is $x \times 10^3 \text{ cm}^{-1}$. The value of x will be:
30. 50% (w/w) H_2SO_4 stock solution has a density of 1.5 g ml^{-1} . The volume (in ml) of stock solution required to prepare a 750 ml solution of 0.4 M H_2SO_4 is :

PART III : MATHEMATICS**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- The equation of the common tangent to the curves $y^2 = 8x$ & $xy = -1$ is:
 (A) $x - 2y + 8 = 0$ (B) $x - y + 2 = 0$ (C) $y - 2x + 1 = 0$ (D) $y - 3x + 2 = 0$
- Volume of parallelepiped with diagonals of three non-parallel adjacent faces along $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$ & $\hat{k} + \hat{i}$ will be:
 (A) 1 (B) 2 (C) 3 (D) None of these
- In a survey of 200 passengers, It was found that 49% liked to travel on bus, 53% liked flight and 62% liked to travel on train, Also, 27% liked to travel on both bus and flight, 29% liked to travel on both flight and train and 28% liked to travel on both train and bus, 5% of these liked none of these transport. The number of passengers who liked at least two of the mode of transports are:
 (A) 120 (B) 108 (C) 124 (D) 104
- Find the image of the point $(0, -1, -3)$ in the plane $3x - y + 4z = 2$.
 (A) $(3, -2, 1)$ (B) $(3, 2, -1)$ (C) $(-3, -2, -1)$ (D) None of these
- The converse of $(x \wedge (\sim y)) \Rightarrow k$ is:
 (A) $\sim k \Rightarrow (\sim x \vee y)$ (B) $k \Rightarrow (\sim x \wedge \sim y)$
 (C) $(\sim x \vee y) \Rightarrow \sim k$ (D) None of these
- For the observation $x_1, x_2, \dots, x_n$; $\sum_{i=1}^n (x_i + 1)^2 = 9n$ & $\sum_{i=1}^n (x_i - 1)^2 = 5n$, the standard deviation of given data is _____.
 (A) $\sqrt{10}$ (B) $\sqrt{5}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$
- The area of quadrilateral formed by the tangents at the ends of latus rectum of the ellipse $9x^2 + 16y^2 = 144$ is:
 (A) $\frac{128}{\sqrt{7}}$ (B) $\frac{64}{7}$ (C) $\frac{2\sqrt{7}}{3}$ (D) $\frac{16\sqrt{7}}{11}$
- The equation of the curve satisfying the differential equation $y_2(x^2 + 1) = 2xy_1$ passing through point $(3, 0)$ and having slope of tangent at $x = 0$ as 2 is:
 (A) $y = \frac{2}{3}x^3 - 4x - 6$ (B) $3y = x^3 - 12x - 3$
 (C) $y = \frac{2}{3}x^3 + 2x - 24$ (D) $3y = \frac{2}{3}x^3 + x - 21$
- The system of linear equation $x + y + z = 6$, $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ has infinite solutions for:
 (A) $\lambda = 4, \mu = 10$ (B) $\lambda = 5, \mu = 5$ (C) $\lambda = 3, \mu = 10$ (D) $\lambda = 10, \mu = 3$

10. The maximum value of the function $f(\theta) = \begin{vmatrix} 1 & 4\cos\theta & 1 \\ \sin\theta & 1 & 4\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$ is:
- (A) 14 (B) 17 (C) $\sqrt{17}$ (D) $\sqrt{14}$
11. Let f and g be differentiable functions satisfying $g'(a) = 2$, $g(a) = b$ & $f \circ g = I$ (Identity function) then $f'(b)$ is equal to:
- (A) $1/2$ (B) 2 (C) $2/3$ (D) None of these
12. If $z^2 - z + 1 = 0$, then value of $\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{24} + \frac{1}{z^{24}}\right)^2$ is equal to:
- (A) 24 (B) 32 (C) 48 (D) None of these
13. If p and q are roots of $ax^2 + bx + c$, then $\lim_{x \rightarrow p} (1 + ax^2 + bx + c)^{\frac{1}{x-p}}$ is :
- (A) $\ln|a(p-q)|$ (B) $e^{a(p-q)}$ (C) $e^{a(q-p)}$ (D) 1
14. Evaluate the value of $\cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right\}$; $x \in \left[\frac{1}{2}, 1\right]$
- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$
15. Let $f(x) = px^2 + qx + r$ be a polynomial function of second degree. If $f(1) = f(-1)$ & $f'(p)$, $f'(q)$, $f'(r)$ are in A.P., then :
- (A) $p = q = r$ (B) p, q, r are in H.P.
(C) p, q, r are in A.P. (D) p, q, r are in G.P.
16. In a parabola $y^2 = 4ax$, if P_1 & P_2 are lengths of perpendicular chords drawn through vertex then $(P_1P_2)^{4/3} = ?$
- (A) $16a^2(P_1^{3/2} + P_2^{3/2})$ (B) $16a^2(P_1^{2/3} + P_2^{2/3})$
(C) $4a^2(P_1^{3/2} + P_2^{3/2})$ (D) $4a^2(P_1^{2/3} + P_2^{2/3})$
17. If $a_1 = \frac{1}{2 \times 5}$, $a_2 = \frac{1}{5 \times 8}$, $a_3 = \frac{1}{8 \times 11}$, and so on then $a_1 + a_2 + a_3 + \dots + a_{100}$ is :
- (A) $\frac{25}{151}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{111}{55}$
18. The sum of integral values of x satisfying $|6x^2 - 5x + 1| = 5x - 6x^2 - 1$ is:
- (A) 5 (B) 3 (C) 2 (D) None of these
19. The value of the integral $\int_0^{80\pi} \frac{dx}{1 + e^{\tan x}}$ is:
- (A) 50π (B) 40π (C) 20π (D) 10π
20. If $\int x \log\left(1 + \frac{1}{x}\right) dx = f(x) \log(x+1) + g(x) \log x^2 + Lx + C$, then:
- (A) $f(x) = \frac{x^2}{2}$ (B) $g(x) = \log x$ (C) $L = 1$ (D) $L = \frac{1}{2}$

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. Find all possible arrangements that can be made by taking 4 letters of the word "VIDYAMANDIR"
22. The positive value of K for which co-efficient of x^4 in the expansion of $(1+x^2)^K$ is 6 is.
23. Number of points of discontinuity of $f(x)=[2x^2]$ in $[1,2)$ is, where $[x]$ denotes greatest integer less than or equal to x .
24. A pair of fair dice is thrown independently 3 times. The probability of getting a score of 7 exactly twice is P . Find $72P$.
25. The distance between $x - \frac{y}{2} + \frac{3}{2} = 0$ & $2x - y + 4 = 0$ is α , and distance between $x - \frac{y}{2} + \frac{3}{2} = 0$ & $3x - \frac{3y}{2} + \frac{15}{2} = 0$ is β , then find $\alpha^2 + \beta^2$.
26. If the domain of inverse of $f(x) = x^2 - 4x + 5$, $x \leq 2$ is $[a, \infty)$ then find the value of ' a '.
27. Let $f(x) = \sqrt{x+9}$. If a point ' C ' satisfies the mean value theorem for the function on the interval $[0, 7]$. Find $4C$.
28. If $2ab^2 - 4ab = 6b^2$, $a > 0$, $b > 0$ then find the minimum value of $\frac{ab}{6}$.
29. Volume of parallelepiped (A, B, C) is $A = (5, 6, 1)$, $B = (0, 2, 3)$, $C = (3, 4, 5)$
30. If the area between $y = x^4$ & $y = \sqrt{x}$ is K . Find $15K$.